

MANDEEP KHATRI

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EDUCATION

Alabama A&M University

Expected Graduation: December 2027

Bachelor of Science in Computer Science, **GPA: 3.80/4.0**

Huntsville, AL

- Awards & Honors:** Dean's List, AAMU Merit Full Scholarship, EICOP 2025 Finalist
- Relevant Coursework:** Data Structures & Algorithms, Intro to programming I and II (Python), Intro to Java Programming, Intro to Digital Logic Design, Calculus I & II

SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, SQL (PostgreSQL, MySQL), HTML, CSS

Frameworks & Tools: Flask, React, Angular, Node.js, Streamlit, LangChain, FAISS, Express, Django, PyMuPDF, Docker, Agile/Scrum, SQLite, SDLC, REST APIs, MQTT, PyTest, JMeter, Spring MVC, CI/CD

AI/ML & Cloud: AWS, GCP (Vertex AI, Gemini API), TensorFlow, PyTorch, NLP, Deep Learning, RAG, Prompt Engineering, Hugging Face Transformers, OpenCV, MLOps (MLflow, Kubeflow)

CS Fundamentals: OOP, Data Structures & Algorithms, Complexity Analysis, Operating Systems, Database Systems

CERTIFICATIONS

Google Cloud Certified – Associate Cloud Engineer

July 2025

WORK EXPERIENCE

Academic Teaching Assistant

January 2025 – Present

Alabama A&M University

Huntsville, AL

- Assist instruction for **80+ freshmen** in **C++, Python, & Calculus I**; prepared course materials, guided labs, and graded assignments, contributing to a **15%** improvement in exam performance.
- Conduct **LeetCode-style** workshops, **live coding demos**, & project-based labs using GitHub Classroom, & **algorithm visualization tools**, mentoring students in collaborative coding & technical problem-solving.

AAMUTeach Program

January 2025 – May 2025

Alabama A&M University

Huntsville, AL

- Led STEM instruction for **70+ students**, introducing interactive classes and projects in **Scratch, Python, and Arduino robotics** to teach programming logic, sensor integration, and algorithm design.
- Developed and delivered **15+ structured modules** using Google Classroom and **hands-on coding labs**, raising assessment scores by **20%** and mentoring student teams in technical problem-solving.

Cloud Computing Intern

January 2023 – March 2023

Verisk

Kathmandu, Nepal

- Built a cloud-native IoT monitoring system with **Python & C++**, deploying microservices on **AWS EC2** & integrating MQTT protocol for device communication; boosted real-time data ingestion & processing efficiency by **30%**.
- Developed a responsive analytics dashboard with **HTML, CSS, & JavaScript**, backed by **Flask/Express RESTful APIs** for live sensor data; reduced user query and visualization time by **20%**.
- Conducted profiling and load testing with **PyTest, JMeter, & Dockerized** containers, resolving bottlenecks & lowering system latency by **25%**, while improving fault tolerance & aligning development with **Agile/SDLC** practices.

PROJECTS

Smart To-Do List | Flask, SQLite, JavaScript, HTML/CSS, Docker, CI/CD

August 2024

- Built a fullstack task manager app using **Flask/Django (Python) + SQLite with CRUD**, prioritization, deadlines, search/filter, and authentication, supporting **100+** persistent tasks.
- Developed **Angular frontend** with **JavaScript, HTML5, CSS3**, integrated Notification API, and automated deployment via **Docker + CI/CD pipelines**, and gained exposure to **Spring MVC** for backend scalability.

LLM-Powered Course Q&A Chatbot | Python, Gemini API, LangChain, FAISS, Streamlit

February 2024

- Built a Q&A chatbot using **Python, Gemini 1.5 Pro API (via Vertex AI)**, and **LangChain**, processing **100+** pages of course PDFs with **PyMuPDF**, text splitting, and embedding via **Gemini**; reached **85%** manual QA accuracy.
- Implemented **RAG** pipeline with **FAISS** (dense retrieval) and **Streamlit** UI, optimizing prompt templates and reducing avg. response latency by **40%** vs. keyword search baseline.

ACTIVITIES

Google Cloud Career Jumpstart HBCU Fellow

May 2025 – August 2025

HBCU Possibilities Summit

January 2025 – May 2025

Apprenticeship Universe Program - HBCU 20x20

May 2025

TracerFire Cybersecurity Competition - Sandia National Laboratories

September 2024